

Estimating Health Condition Prevalence Among a Statewide Cohort with Recent Homelessness or Incarceration



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ABSTRACT

BACKGROUND: Public health data systems have limited ability to provide timely, population-level information for people with severe and multiple disadvantages, such as individuals with recent homelessness or incarceration.

OBJECTIVE: To generate prevalence estimates for physical health, mental health, and substance use conditions in a statewide cohort that included individuals with recent incarceration or homelessness.

DESIGN: This observational cohort analysis was completed in July 2025 and used linked statewide electronic health record (EHR) and administrative data through the Minnesota Electronic Health Record Consortium (MNEHRC) and its Health Trends Across Communities project.

PARTICIPANTS: Adults with an encounter at a MNEHRC-participating health system between 2021–2023.

MAIN MEASURES: Statewide directly standardized, age and sex-adjusted, prevalence rates of 22 health conditions chosen by public health, healthcare, and research leaders in Minnesota, stratified by recent homelessness, jail incarceration, or prison incarceration.

KEY RESULTS: This cohort included 4,362,645 individuals (53% female, 73% white), including 20,139 individuals with recent homelessness, 51,470 individuals with recent jail incarceration, and 4,889 individuals with recent prison incarceration. Individuals with recent homelessness or jail or prison incarceration had a higher prevalence of asthma (14.9%, 9.6%, 10.1% respectively, vs. 7.1%) and COPD (10.5%, 6.1%, 5.5%, respectively, vs. 3.0%) compared to the general population. Individuals with recent homelessness had the highest rates of mental health disorders compared to

other included groups; recently homeless Black individuals had the highest recorded rates of psychotic disorder diagnoses (18.7%) across racial and ethnic groups experiencing homelessness. Substance use disorders among individuals with recent homelessness or jail or prison incarceration, including opioid use disorder (13.9%, 10.6%, and 13.6%, respectively, vs. 1.4%), were higher compared to the general population.

CONCLUSIONS: Our findings highlight widespread disparities among people with recent homelessness or incarceration, particularly related to mental health and substance use conditions. Payment and delivery models that account for high levels of co-occurring health and social complexity are needed.

KEY WORDS: homelessness; incarceration; electronic health record; chronic disease

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INTRODUCTION

Monitoring the prevalence of health conditions in communities is critical for informing healthcare and public health policy and delivery.^{1–3} However, health condition prevalence data in the United States (US) are primarily based on survey methods, which often have a 2 to 3 year lag and lack the geographic granularity needed by local and regional healthcare systems and public health agencies.^{4,5} These limitations are particularly pronounced for populations with severe and multiple disadvantages, like individuals who have recently experienced homelessness or incarceration. People with recent homelessness or incarceration have a high burden of physical health,⁶ mental

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health,⁷ and substance use conditions^{7,8} compared with the general population, but they are typically absent from public health data sources.^{9,10} Robust data systems that include these populations are needed to inform policy and ultimately overcome substantial barriers to critically needed medical and behavioral health services, like vaccinations and medication for opioid use disorder.^{11,12}

The Minnesota Electronic Health Record Consortium (MNEHRC) is a unique healthcare and public health collaboration that provides regularly updated health data on over 5.8 million Minnesotans.¹³ Through collaboration between healthcare systems, the Minnesota Department of Health, and other state and local agencies, the MNEHRC links electronic health record (EHR) data with statewide information on homelessness, incarceration, Medicaid enrollment, and mortality. Recently, our collaboration expanded previous work on COVID-19 to estimate and map health condition prevalence at the census tract level in Minnesota. Local public health agencies selected the initial health indicators and prioritized data that would be most likely to impact their agencies' work over the next several years.

In this analysis using direct standardization, we provide prevalence estimates of physical health, mental health, and substance use conditions for a statewide population of people with recent homelessness or incarceration compared with the general population. To our knowledge, these are the first EHR-based statewide prevalence estimates among these populations in the US and could serve as a potential tool for future health policy and delivery changes.

METHODS

Cohort Design and Setting

MNEHRC includes 11 Minnesota healthcare systems that contribute data: Allina Health, CentraCare, Children's Minnesota, Essentia Health, HealthPartners, Hennepin Healthcare, M Health Fairview, Mayo Clinic, Minneapolis VA, Sanford Health, and North Memorial Health. Each of the 11 healthcare care systems maintains their EHR data in the Observational Medical Outcomes Partnership (OMOP) common data model. Centralized R code is distributed to sites to query their database and generate descriptive data. We included any individual 18 and older with a Minnesota address in their health record and an encounter at a participating healthcare system (i.e., any outpatient, telehealth, emergency department, or inpatient visit) in the past three years (2021–2023, $n=4,362,645$). Records were deduplicated so that patients seen at multiple health systems had their condition(s) counted only once.

This project was funded by the Minnesota Department of Health and was determined to be a public health

surveillance project that did not require review from institutional review boards.

Physical Health, Mental Health, and Substance Use Conditions

Records in the OMOP common data model use concept IDs that represent conditions in EHR data vocabularies like International Classification of Diseases (ICD) and Systematized Nomenclature of Medicine (SNOMED). We created codesets of OMOP concept IDs that identify each condition. We first converted ICD-based diagnoses from the CMS Chronic Condition Warehouse¹⁴ to create a base set of OMOP concept IDs. We then used a combined metadata file from all sites to identify additional concept IDs related to the conditions of interest. After each codeset was created, a clinician reviewed codes for final inclusion or exclusion. Codes where there was uncertainty were adjudicated by a second clinician with expertise in OMOP and health services research. All codesets are available upon request. Hypertension includes a diagnosis and/or two outpatient blood pressure measurements $\geq 140/90$. Diabetes mellitus includes a diagnosis and/or an A1c $\geq 6.5\%$. We describe substance-related conditions in this paper as substance use disorders because the diagnosis-based codesets generally represent problematic substance use and not use alone, though we did not have formal diagnostic assessment data available.

Homelessness and Incarceration Data

Data on homelessness came from the Minnesota Homeless Management Information System which captures information regarding use of homeless services like street outreach, emergency shelter, and supportive housing.¹⁵ Federal housing programs (e.g. Section 8 housing) are not included. Data on incarceration came from records maintained by the Minnesota Department of Corrections on all admissions and discharges in Minnesota county jails and state prisons. Multiple healthcare systems that participate within the MNEHRC provide healthcare in county jails, while no healthcare system provides healthcare in state prisons. Diagnoses that are made during a jail encounter with a participating MNEHRC site are included in the analysis. For this analysis, we used a one-year lookback period for both homelessness and incarceration indexed to the date of the most recent analysis (HTAC version 2025.3, April/May 2025). For 2023, we report on 51,470 individuals of 82,040 who had any jail incarceration event. We report on 4,889 individuals of 11,936 who had a prison incarceration event in 2023. The denominators for jail and prison incarceration is inclusive of people currently incarcerated who would not have had community health care use. We report on 20,139 individuals of 50,184 who had experienced homelessness in 2023. The denominator for recent homelessness is inclusive of children who were not reported in this analysis.

Sociodemographic Characteristics

Each site-specific common data file contained sociodemographic characteristics, including age, sex, race and ethnicity, socioeconomic status, and geography. Race and ethnicity (Hispanic ethnicity, non-Hispanic race including American Indian or Alaska Native, Asian or Pacific Islander, Black, multiracial, White) were determined at the time of an individual's registration and are self-reported characteristics in most health systems.

Statistical Analysis

We performed analyses using R Version 4.3.2 and Stata 15.1. Data were generated at each site using a standardized R script and aggregated at a central data coordinating site using R. To generate directly standardized prevalence rates and 95% confidence intervals, we used Stata's *dstdize* function and adjusted for age categories (i.e., 18–24, 25–34, 35–44, 45–54, 55–64, 65–74, and 75 and older) and sex. We standardized estimates to the study sample, which is largely representative of the Minnesota adult population. Of note, recently incarcerated and homeless populations are nested populations within the general population and samples were not entirely independent though they were treated as such. Individuals younger than 18 or who were missing sex information (0.1%) were removed from the sample prior to analysis. For race and ethnicity estimates, we removed individuals with unknown or missing race and ethnicity information (8.1%).

RESULTS

Cohort Demographic Data

The cohort included 4,362,645 individuals, with the majority female (53%), White (73%). Our cohort was similar to the US Census-reported Minnesota adult population of approximately 4.5 million individuals. Among the total population, 0.5% ($n=20,139$), 1.2% ($n=51,470$), and 0.1% ($n=4,889$) had recently experienced homelessness, jail incarceration, or prison incarceration, respectively. Individuals with recent homelessness were more likely to be younger (age 25–44, 48% vs. 34%), male (51% vs. 47%), and Black (34% vs. 8%) or American Indian/Alaska Native (11% vs. 1%) compared to the general population (Table 1). Similarly, individuals with recent jail or prison incarceration were more likely to be younger (age 25–44, 61% and 64%, respectively, vs. 34%), male (74% and 88%, respectively, vs. 47%), and Black (23% and 31%, respectively, vs. 8%) compared to the general population (Table 1).

Physical Health, Mental Health, and Substance use Conditions Among People with Recent Homelessness

Adjusted for age and sex, prevalence of asthma (14.9% [95% CI, 14.1–15.6]) and chronic obstructive pulmonary disease (COPD) (10.5% [95% CI, 9.6–11.5]), were higher among individuals with recent homelessness, compared to the general population (Asthma: 7.1% [95% CI, 7.1–7.1] and COPD: 3.0% [95% CI, 3.0–3.0]). Rates of heart failure (6.6% [95% CI, 5.8–7.4] vs. 3.7% [95% CI, 3.6–3.7]),

Table 1 Sociodemographic Characteristics of Statewide Cohort—Minnesota, 2023

	General Population	Recent Homelessness	Recent Jail Incarceration	Recent Prison Incarceration
N	4,362,645	20,139	51,470	4,889
Sex				
Male	47% (2,031,043)	51% (10,348)	74% (38,138)	88% (4,315)
Female	53% (2,331,602)	49% (9,791)	26% (13,332)	12% (574)
Age				
18–24	11% (482,500)	14% (2,883)	13% (6,566)	8% (395)
25–34	17% (725,266)	23% (4,557)	31% (15,965)	30% (1,485)
35–44	17% (743,904)	25% (5,063)	30% (15,365)	34% (1,663)
45–54	14% (625,356)	17% (3,492)	15% (7,774)	16% (790)
55–64	16% (693,514)	15% (3,104)	9% (4,405)	8% (412)
65–74	14% (616,292)	4% (904)	2% (1,224)	2% (121)
75 and older	11% (475,813)	1% (136)	0.3% (171)	0.5% (23)
Race and Ethnicity				
American Indian/Alaska Native	1% (41,636)	11% (2,149)	7% (3,558)	8% (413)
Asian or Pacific Islander	5% (205,403)	2% (384)	3% (1,492)	2% (104)
Black	8% (348,564)	34% (6,757)	23% (11,944)	31% (1,516)
Hispanic	5% (225,295)	5% (1,080)	7% (3,525)	7% (320)
White	73% (3,186,964)	41% (8,180)	54% (27,717)	47% (2,279)
Other/Missing/Unknown	8% (354,783)	8% (1,589)	6% (3,234)	5% (257)

acute myocardial infarction (AMI) (4.9% [95% CI, 4.2, 5.6] vs. 2.1% [95% CI, 2.1–2.1]), and type II diabetes (14.9% [95% CI, 13.7–16.1] vs. 9.7% [95% CI, 9.7–9.8]) were also higher compared to the general population (Table 2).

We identified substantial differences in mental health and substance use condition prevalence between homeless and general populations. Depression (37.4% [95% CI, 36.2–38.6] vs. 17.2% [95% CI, 17.2–17.3]), anxiety (43.7% [95% CI, 42.4–44.9] vs. 22.5% [95% CI, 22.4–22.5]), post-traumatic stress disorder (PTSD) (13.8% [95% CI, 13.1–14.5] vs. 2.1% [95% CI, 2.1–2.1]), and psychotic disorders (17.8% [95% CI, 17.0–18.7] vs. 1.8% [95% CI, 1.7–1.8]) were higher among people with recent homelessness compared with the general population (Table 2). Similarly, opioid use disorder (13.9% [95% CI, 13.3–14.5] vs. 1.4% [95% CI, 1.4–1.4]), alcohol use disorder (21.2% [95% CI, 20.4–21.9] vs. 3.7% [95% CI, 3.7–3.7]), methamphetamine use disorder (15.2% [95% CI, 14.5–15.8] vs. 0.8% [95% CI, 0.8–0.8]), cocaine use disorder (5.5% [95% CI, 5.1–6.0] vs. 0.3% [95% CI, 0.3–0.3]), and cannabis use disorder (12.1% [95% CI, 11.6–12.6] vs. 1.6% [95% CI, 1.5–1.6]) were more common among people with recent homelessness compared with the general population (Table 2).

Physical Health, Mental Health, and Substance use Conditions Among People with Recent Jail Or Prison Incarceration

Similar to health condition prevalence in individuals with recent homelessness, those with recent jail incarceration had a higher prevalence of several physical health, mental health and substance use conditions. Asthma (9.6% [95% CI, 9.2–10.1] vs. 7.1% [95% CI, 7.1–7.1]), depression (30.6% [95% CI, 30.0–31.3] vs. 17.2% [95% CI, 17.2–17.3]), anxiety (37.3% [95% CI, 36.5–38.1] vs. 22.5% [95% CI, 22.4–22.5]), psychotic disorders (10.8% [95% CI, 10.3–11.3] vs. 1.8% [95% CI, 1.7–1.8]), opioid use disorder (10.6% [95% CI, 10.2–11.0] vs. 1.4% [95% CI, 1.4–1.4]), alcohol use disorder (21.0% [95% CI, 20.4–21.7] vs. 3.7% [95% CI, 3.7–3.7]), methamphetamine use disorder (12.2% [95% CI, 11.9–12.6] vs. 0.8% [95% CI, 0.8–0.8]), cocaine use disorder (3.3% [95% CI, 3.0–3.5] vs. 0.3% [95% CI, 0.3–0.3]), and cannabis use disorder (9.1% [95% CI, 8.8–9.4] vs. 1.6% [95% CI, 1.5–1.6]) were higher among people with recent jail incarceration compared with the general population. Prevalence of peripheral vascular disease (2.6% vs. 2.4%), CAD/ischemic heart disease (5.9% vs. 5.4%), and stroke (2.8% vs. 2.3%) were similar among individuals with recent jail incarceration compared with the general population (Table 2).

Table 2 Prevalence of Physical Health, Mental Health, and Substance Use Conditions Among Statewide Cohort—Minnesota, 2023

Condition	General Population		Homeless		Recent Jail Incarceration		Recent Prison Incarceration	
	Prevalence (%)	95% CI	Prevalence (%)	95% CI	Prevalence (%)	95% CI	Prevalence (%)	95% CI
Physical Health								
Asthma	7.1%	7.1, 7.1	14.9%	14.1–15.6	9.6%	9.2, 10.1	10.1%	7.7, 12.5
COPD	3.0%	3.0, 3.0	10.5%	9.6, 11.5	6.1%	5.6, 6.6	5.5%	3.2, 7.8
Hypertension	33.4%	33.4, 33.5	37.3%	36.3, 38.4	37.8%	37.0, 38.7	26.3%	23.4, 29.1
Hyperlipidemia	23.8%	23.7, 23.8	19.9%	18.7, 21.1	17.4%	16.7, 18.2	13.0%	10.5, 15.5
Diabetes mellitus, type 2	9.7%	9.7, 9.8	14.9%	13.7, 16.1	9.4%	8.8, 10.0	8.1%	6.2, 9.9
CAD/Ischemic heart disease	5.5%	5.5, 5.5	7.6%	6.8, 8.5	5.9%	5.3, 6.4	5.3%	4.3, 6.4
Heart failure	3.7%	3.6, 3.7	6.6%	5.8, 7.4	4.1%	3.6, 4.6	5.0%	2.8, 7.3
Stroke	2.5%	2.5, 2.5	4.4%	3.7, 5.1	2.8%	2.3, 3.2	2.6%	2.1, 3.2
AMI	2.1%	2.1, 2.1	4.9%	4.2, 5.6	3.2%	2.7, 3.7	3.2%	2.0, 4.4
Peripheral vascular disease	2.4%	2.4, 2.4	4.2%	3.4, 5.0	2.6%	2.1, 3.0	2.4%	0.3, 4.5
Mental Health								
Depression	17.2%	17.2, 17.3	37.4%	36.2, 38.6	30.6%	30.0, 31.3	24.7%	21.9, 27.4
Anxiety	22.5%	22.4, 22.5	43.7%	42.4, 44.9	37.3%	36.5, 38.1	30.2%	28.0, 32.3
PTSD	2.1%	2.1, 2.1	13.8%	13.1, 14.5	9.3%	8.9, 9.7	8.5%	7.2, 9.9
Bipolar disorder	1.6%	1.6, 1.6	11.8%	11.2, 12.3	8.1%	7.7, 8.5	8.0%	5.7, 10.2
Psychotic disorders	1.8%	1.7, 1.8	17.8%	17.0, 18.7	10.8%	10.3, 11.3	7.9%	6.8, 9.1
Suicide attempt/ideation	1.6%	1.6, 1.6	12.2%	11.6, 12.7	8.9%	8.5, 9.3	7.2%	6.0, 8.4
Substance Use								
Opioid	1.4%	1.4, 1.4	13.9%	13.3, 14.5	10.6%	10.2, 11.0	13.6%	11.2, 15.9
Alcohol	3.7%	3.7, 3.7	21.2%	20.4, 21.9	21.0%	20.4, 21.7	13.6%	11.1, 16.0
Hallucinogens	0.05%	0.05, 0.05	0.4%	0.3, 0.4	0.4%	0.3, 0.5	0.2%	0.1, 0.3
Methamphetamine	0.8%	0.8, 0.8	15.2%	14.5, 15.8	12.2%	11.9, 12.6	17.3%	14.8, 19.8
Cocaine	0.3%	0.3, 0.3	5.5%	5.1, 6.0	3.3%	3.0, 3.5	4.1%	2.0, 6.1
Cannabis	1.6%	1.5, 1.6	12.1%	11.6, 12.6	9.1%	8.8, 9.4	6.7%	5.5, 8.0

Estimates adjusted for age and sex

Abbreviation(s): COPD, chronic obstructive pulmonary disease. CAD, coronary artery disease. AMI, acute myocardial infarction. PTSD, post-traumatic stress disorder

Individuals with recent prison incarceration had higher rates of asthma (10.1% [95% CI, 7.7–12.5] and COPD (5.5% [95% CI, 3.2–7.8] compared to the general population (Asthma: 7.1% [95% CI, 7.1–7.1] and COPD: 3.0% [95% CI, 3.0–3.0]) and similar rates of other physical health conditions. Those with recent prison incarceration had higher prevalence of mental health and substance use conditions, including, depression (24.7% [95% CI, 21.9–27.4] vs. 17.2% [95% CI, 17.2–17.3]), anxiety (30.2% [95% CI, 28.0–32.3] vs. 22.5% [95% CI, 22.4–22.5]), bipolar disorder (8.0% [95% CI, 5.7–10.2] vs. 1.6% [95% CI, 1.6–1.6]), opioid use disorder (13.6% [95% CI, 11.2–15.0] vs. 1.4% [95% CI, 1.4–1.4]), and methamphetamine use disorder (17.3% [95% CI, 14.8–19.8] vs. 0.8% [95% CI, 0.8–0.8]), compared to the general population (Table 2).

Racial and Ethnic Disparities Among People with Recent Homelessness

We identified important racial differences in health condition prevalence among individuals with recent homelessness. Black individuals with recent homelessness had the highest proportion of asthma (18.0% [95% CI, 16.0–20.0]) and hypertension (36.8% [95% CI, 34.5–39.1]) among people with recent homelessness. Type 2 diabetes was higher among Minnesotans with recent homelessness who identified as Hispanic (17.8% [95% CI, 14.4–21.2]), Asian/Pacific Islander (17.0% [95% CI, 11.8–22.1]), Black (14.7% [95% CI, 12.7–16.7]), or Native American/Alaska Native (15.0% [95% CI, 12.7–17.2]) compared to White individuals (11.6% [95% CI, 10.6–12.6]) with recent homelessness (Table 3).

White Minnesotans with recent homelessness had a greater proportion of depression (40.5% [95% CI, 39.2–41.8]) and anxiety (48.7% [95% CI, 47.4–50.1]) compared to Black (31.9% [95% CI, 29.9–34.0]; 33.2% [95% CI, 31.3–35.1], respectively), Native American/Alaska Native (26.6% [95% CI, 24.4–28.9] and 33.3% [95% CI 31.0–35.5], respectively), and Hispanic individuals with recent homelessness (32.4% [95% CI, 26.7–38.1]; 40.5% [95% CI, 34.7–46.2], respectively). Black individuals with recent homelessness had the highest prevalence of psychotic disorders (18.7% [95% CI, 17.9–20.5]) compared with White (16.2% [95% CI, 15.2–17.1]), Native American/Alaska Native (15.0% [95% CI, 13.5–16.6]), Hispanic (14.3% [95% CI, 12.2–16.4]), and Asian/Pacific Islander individuals with recent homelessness (9.2% [95% CI, 6.6–11.9]).

Native American/Alaska Native individuals with recent homelessness had higher rates of opioid use (20.3% [95% CI, 18.4–22.1]) compared to White (13.4%, [95% CI, 12.7–14.2]), Black (11.2% [95% CI, 9.7–12.7]) and Hispanic (10.5% [95% CI, 8.7–12.2]) individuals with recent homelessness (Table 3).

Racial and Ethnic Disparities Among People with Recent Jail or Prison Incarceration

Black individuals with recent jail incarceration had higher rates of asthma (12.7% [95% CI, 11.3–14.2]) compared to White (9.1% [95% CI, 8.5–9.6]), Native American/Alaska Native (8.1% [95% CI, 6.2–10.0]), Hispanic (7.9% [95% CI, 6.3–9.4]), and Asian/Pacific Islander (7.9% [95% CI, 2.6–13.1]) individuals with recent incarceration. Hypertension was highest among White and Hispanic individuals with recent jail incarceration (36.2% [95% CI, 35.3–37.0] and 37.6% [95% CI, 35.0–40.1], respectively) compared to Native Americans (28.9% [95% CI, 25.0–32.0] or Asian/Pacific Islanders (31.9% [95% CI, 28.7–35.0]) (Table 4).

White individuals with recent jail incarceration had higher rates of depression (32.9% [95% CI, 32.0–33.7]) and anxiety (40.2% [95% CI, 39.4–41.1]) when compared to other racial and ethnic groups. Black individuals with recent jail incarceration had the highest prevalence of psychotic disorders (14.2% [95% CI 12.6–15.9]) compared to other racial and ethnic groups with recent jail incarceration. Native American/Alaska Native individuals with recent jail incarceration had the highest prevalence of opioid (17.2% [95% CI, 14.9–19.5]) and methamphetamine use disorder (18.4% [95% CI, 16.3–20.4]) compared to all other racial and ethnic groups with recent incarceration (Table 4).

White individuals with recent prison incarceration had higher prevalence of COPD (7.1%, [95% CI, 3.7–10.6]) compared to Black (1.5% [95% CI, 0.6–2.5]), Native American/Alaska Native (1.0% [0–2.5]), and Hispanic (0.4% [95% CI, 0–1.1]) individuals with recent prison incarceration. Black individuals with recent prison incarceration had highest rates of psychotic disorders (12.7% [95% CI, 8.6–16.9]) compared to individuals of all other races with recent prison incarceration. Finally, Black individuals with recent incarceration (15.6%, [95% CI, 11.5–19.8]) had higher rates of cocaine use disorder compared with White (6.2% [95% CI, 4.5–7.9]), Native American/Alaska Native (6.6% [95% CI, 2.3–10.9]), Hispanic (9.1% [95% CI, 5.1–13.1]), and Asian/Pacific Islander (4.5% [95% CI, 0.8–8.2]) individuals with recent prison incarceration (Table 5).

DISCUSSION

Through linkage of statewide EHR and administrative data, we produced timely estimates of physical health, mental health, and substance use conditions among Minnesotans with recent homelessness or incarceration. Our findings highlight the substantial severe and multiple disadvantages faced by these populations; policy reforms are needed to ensure that necessary medical and social support can be accessed by these populations and that such care is financially sustainable. Our findings were produced through a

Table 3 Prevalence of Health Conditions Among Individuals with Recent Homelessness by Race and Ethnicity—Minnesota, 2023

Condition	White		Black		Native American/ Alaska Native		Hispanic		Asian/Pacific Islander	
	Prevalence (%)	95% CI	Prevalence (%)	95% CI	Prevalence (%)	95% CI	Prevalence (%)	95% CI	Prevalence (%)	95% CI
Physical Health										
Asthma	14.4%	13.4, 15.4	18.0%	16.0, 20.0	10.4%	8.8, 11.9	13.9%	8.6, 19.3	10.1%	6.6, 13.6
COPD	11.7%	10.7, 12.8	7.0%	5.5, 8.5	6.4%	4.7, 8.1	6.5%	1.3, 11.7	7.0%	2.1, 11.8
Hypertension	35.4%	34.2, 36.5	36.8%	34.5, 39.1	24.5%	22.0, 27.0	33.8%	31.4, 36.3	28.8%	23.4, 34.1
Hyperlipidemia	18.7%	17.5, 19.9	17.8%	15.6, 20.0	11.6%	9.6, 13.5	15.4%	9.9, 20.9	18.5%	13.0, 24.0
Diabetes mellitus, type 2	11.6%	10.6, 12.6	14.7%	12.7, 16.7	15.0%	12.7, 17.2	17.8%	14.4, 21.2	17.0%	11.8, 22.1
CAD/Ischemic heart disease	7.1%	6.2, 8.0	7.7%	6.1, 9.4	6.0%	4.4, 7.6	3.8%	2.4, 5.3	6.4%	3.2, 9.5
Heart failure	6.0%	5.2, 6.8	6.7%	5.4, 8.1	3.9%	2.7, 5.1	2.7%	1.4, 3.9	5.7%	2.6, 8.7
Stroke	4.1%	3.3, 4.8	4.5%	3.1, 5.9	4.5%	3.0, 5.9	3.0%	1.7, 4.3	4.0%	1.2, 6.9
AMI	4.8%	4.0, 5.5	5.5%	4.2, 6.8	3.9%	2.6, 5.1	3.4%	2.0, 4.8	3.7%	0.9, 6.5
Peripheral vascular disease	3.3%	2.6, 3.9	2.6%	1.7, 3.5	2.6%	1.5, 3.7	1.5%	0.6, 2.5	1.6%	0, 3.7
Mental Health										
Depression	40.5%	39.2, 41.8	31.9%	29.9, 34.0	26.6%	24.4, 28.9	32.4%	26.7, 38.1	33.7%	28.0, 39.5
Anxiety	48.7%	47.4, 50.1	33.2%	31.3, 35.1	33.3%	31.0, 35.5	40.5%	34.7, 46.2	34.8%	28.7, 40.8
PTSD	14.9%	14.1, 15.7	11.4%	10.4, 12.3	11.2%	9.9, 12.5	12.0%	10.2, 13.8	11.1%	7.7, 14.5
Bipolar disorder	14.2%	13.3, 15.0	10.0%	9.0, 11.0	6.0%	4.9, 7.0	9.7%	8.0, 11.4	6.3%	4.3, 8.3
Psychotic disorders	16.2%	15.2, 17.1	18.7%	17.9, 20.5	15.0%	13.5, 16.6	14.3%	12.2, 16.4	15.5%	11.9, 19.2
Suicide attempt/ideation	14.2%	13.3, 15.0	9.3%	8.4, 10.1	9.6%	8.2, 11.0	10.4%	8.7, 12.1	9.2%	6.6, 11.9
Substance Use										
Opioid	13.4%	12.7, 14.2	11.2%	9.7, 12.7	20.3%	18.4, 22.1	10.5%	8.7, 12.2	16.4%	12.3, 20.5
Alcohol	21.7%	20.6, 22.8	18.7%	17.4, 20.1	24.2%	22.2, 26.3	17.1%	14.9, 19.3	13.0%	9.1, 16.8
Hallucinogens	0.3%	0.2, 0.4	0.7%	0.5, 0.8	0.1%	0, 0.2	0.2%	0, 0.4	0.0%	0, 0
Methamphetamine	18.6%	17.7, 19.4	6.9%	6.3, 7.6	18.4%	16.7, 20.0	13.6%	11.7, 15.5	16.4%	12.8, 20.0
Cocaine	3.4%	2.9, 3.9	10.5%	9.0, 12.1	3.1%	2.3, 3.9	3.3%	2.1, 4.5	2.1%	0.9, 3.4
Cannabis	12.3%	11.6, 13.1	12.7%	11.7, 13.7	10.0%	8.5, 11.5	9.5%	7.9, 11.2	6.1%	4.0, 8.3

Estimates adjusted for age and sex

Abbreviation(s): COPD, chronic obstructive pulmonary disease. CAD, coronary artery disease. AMI, acute myocardial infarction. PTSD, post-traumatic stress disorder

linkage process that is scalable and could be replicated in additional jurisdictions to produce actionable data that drives care for people with severe and multiple disadvantages.

The prevalence of health conditions by race and ethnicity in individuals with recent homelessness or incarceration varied in important ways. For example, the prevalence of mental depression and anxiety were higher among White individuals with recent homelessness or jail incarceration compared with other racial and ethnic groups, while psychotic disorders were higher among Black individuals with recent homelessness or incarceration. Our estimates of physical health conditions among individuals with recent incarceration,

such as COPD, hypertension, diabetes, and ischemic heart disease, were higher than similar analyses of claims from Hennepin County but identified similar racial disparities and trends.¹⁸ Finally, we found notably high rates of psychotic disorders, opioid use disorder, and methamphetamine use disorder among Native American Minnesotans, which has been previously reported.^{19,20}

We found EHR-based depression prevalence among individuals with recent jail incarceration was higher compared with self-reported national estimates of people with recent criminal justice involvement (30.6% vs. 13.9%)²¹ but lower compared with estimates using linked healthcare claims data among people on

Table 4 Prevalence of Health Conditions among Individuals with Recent Jail Incarceration by Race and Ethnicity—Minnesota, 2023

Condition	White		Black		Native American/Alaska Native		Hispanic		Asian/Pacific Islander	
	Prevalence (%)	95% CI	Prevalence (%)	95% CI	Prevalence (%)	95% CI	Prevalence (%)	95% CI	Prevalence (%)	95% CI
Physical Health										
Asthma	9.1%	8.5, 9.6	12.7%	11.3, 14.2	8.1%	6.2, 10.0	7.9%	6.3, 9.4	7.9%	2.6, 13.1
COPD	6.1%	5.5, 6.7	6.0%	4.2, 7.7	4.3%	2.4, 6.2	2.8%	1.4, 4.3	0.3%	0, 0.6
Hypertension	36.2%	35.3, 37.0	33.7%	31.8, 35.5	28.9%	25.0, 32.9	37.6%	35.0, 40.1	31.9%	28.7, 35.0
Hyperlipidemia	17.0%	16.3, 17.8	15.0%	12.9, 17.1	10.8%	7.5, 14.1	15.5%	10.0, 21.0	14.4%	8.2, 20.6
Diabetes mellitus, type 2	7.9%	7.3, 8.6	11.3%	9.8, 12.8	14.8%	11.2, 18.4	12.5%	10.2, 14.8	10.6%	7.9, 13.3
CAD/Ischemic heart disease	5.1%	4.6, 5.7	5.5%	3.7, 7.2	4.4%	2.2, 6.6	3.2%	1.8, 4.6	1.7%	0.2, 3.1
Heart failure	3.8%	3.4, 4.3	3.9%	2.8, 5.0	3.3%	2.1, 4.4	1.8%	1.0, 2.6	2.4%	0.9, 4.0
Stroke	2.3%	2.0, 2.7	2.5%	1.6, 3.4	3.0%	2.0, 4.1	1.3%	0.6, 2.0	2.0%	0.5, 3.5
AMI	2.6%	2.2, 3.0	4.1%	2.7, 5.6	2.9%	1.8, 4.0	1.6%	0.7, 2.4	1.5%	0.04, 2.9
Peripheral vascular disease	2.3%	1.9, 2.7	1.6%	0.8, 2.5	3.6%	1.1, 6.1	1.5%	0.5, 2.5	0.2%	0, 0.5
Mental Health										
Depression	32.9%	32.0, 33.7	24.8%	22.8, 26.7	21.6%	19.4, 23.9	23.2%	20.9, 25.4	13.0%	10.8, 15.3
Anxiety	40.2%	39.4, 41.1	28.7%	26.5, 30.8	34.7%	31.8, 37.6	29.0%	26.6, 31.4	19.0%	13.5, 24.5
PTSD	9.5%	9.0, 10.0	9.4%	7.9, 10.9	9.2%	7.4, 11.0	7.3%	5.9, 8.8	4.0%	2.9, 5.1
Bipolar disorder	8.6%	8.1, 9.0	8.2%	6.8, 9.6	4.7%	3.8, 5.6	5.6%	4.3, 6.9	7.1%	2.0, 12.2
Psychotic disorders	9.4%	8.9, 9.9	14.2%	12.6, 15.9	13.1%	10.7, 15.5	8.5%	6.9, 10.1	6.7%	4.8, 8.5
Suicide attempt/ideation	9.6%	9.1, 10.1	7.4%	6.2, 8.6	9.7%	7.8, 11.6	8.0%	6.5, 9.4	6.9%	5.0, 8.9
Substance Use										
Opioid	10.0%	9.6, 10.5	10.7%	9.4, 12.0	17.2%	14.9, 19.5	7.8%	6.4, 9.2	4.8%	3.7, 5.9
Alcohol	22.0%	21.2, 22.7	18.1%	16.3, 20.0	22.5%	19.1, 25.9	18.1%	12.7, 23.5	12.1%	9.0, 15.3
Hallucinogens	0.5%	0.3, 0.6	0.6%	0.5, 0.8	0.1%	0.02, 0.1	0.2%	0.1, 0.4	0.1%	0, 0.2
Methamphetamine	13.6%	13.1, 14.1	6.2%	5.4, 7.1	18.4%	16.3, 20.4	10.9%	9.4, 12.5	8.0%	6.7, 9.3
Cocaine	2.3%	2.0, 2.5	9.4%	7.9, 10.9	2.2%	1.6, 2.9	2.8%	1.8, 3.8	1.4%	0.7, 2.0
Cannabis	9.1%	8.6, 9.5	10.6%	9.8, 11.5	9.0%	7.1, 11.0	6.2%	5.3, 7.1	4.3%	3.2, 5.4

Estimates adjusted for age and sex

Abbreviation(s): COPD, chronic obstructive pulmonary disease. CAD, coronary artery disease. AMI, acute myocardial infarction. PTSD, post-traumatic stress disorder

probation in Hennepin County, Minnesota (30.6% vs. 41.6%).¹⁸ Compared with EHR-based estimates of health conditions among individuals with a history of homelessness in Ohio,⁶ we found lower rates of hypertension (37.3% vs 44.2%), diabetes (14.9% vs. 19.2%), heart failure (6.6% vs. 8.4%), depression (37.4% vs. 48.1%), and alcohol use disorder (21.2% vs. 30.6%). Finally, our estimates of mental health and substance use conditions among people with recent homelessness or incarceration were higher compared with recent systematic reviews assessing

behavioral prevalence in these populations.^{7,22,23} We hypothesize the higher prevalence we identified may be related to improved diagnosis rates in the community compared with individuals who are receiving care during incarceration or because our population, by definition, is a population that is using healthcare and may be sicker at baseline.

Our report offers a path forward for generating prevalence estimates among populations with severe and multiple disadvantages. Because health condition data among people with

Table 5 Prevalence of Health Conditions among Individuals with Recent Prison Incarceration by Race and Ethnicity—Minnesota, 2023

Condition	White		Black		Native American/ Alaska Native		Hispanic		Asian/Pacific Islander	
	Prevalence (%)	95% CI	Prevalence (%)	95% CI	Prevalence (%)	95% CI	Prevalence (%)	95% CI	Prevalence (%)	95% CI
Physical Health										
Asthma	10.2%	7.9, 12.6	16.6%	12.9, 20.2	5.3%	2.4, 8.2	11.1%	4.4, 17.9	2.8%	0.0, 6.3
COPD	7.1%	3.7, 10.6	1.5%	0.6, 2.5	1.0%	0, 2.5	0.4%	0, 1.1	0.0%	0, 0
Hypertension	26.8%	22.7, 31.0	26.8%	19.6, 34.0	18.1%	12.3, 24.0	21.4%	14.7, 28.1	14.5%	8.8, 20.1
Hyperlipidemia	14.0%	10.2, 17.8	13.2%	9.2, 17.2	8.4%	3.5, 13.2	3.0%	1.4, 4.5	7.9%	2.3, 13.5
Diabetes mellitus, type 2	7.7%	4.8, 10.7	10.5%	6.2, 14.8	9.6%	4.6, 14.6	2.7%	1.4, 4.1	4.7%	1.5, 8.1
CAD/Ischemic heart disease	4.5%	3.2, 5.8	3.0%	1.2, 4.9	3.0%	0.4, 5.7	1.2%	0.03, 2.4	3.8%	0, 8.5
Heart failure	5.1%	1.8, 8.4	4.8%	1.7, 7.9	2.7%	0.8, 4.6	0.5%	0, 1.2	0.0%	0, 0
Stroke	1.4%	0.7, 2.1	1.6%	0.6, 2.6	1.4%	0.1, 2.8	0.6%	0, 1.4	0.9%	0, 1.9
AMI	2.6%	1.6, 3.7	2.0%	0.7, 3.3	2.1%	0, 4.1	1.4%	0, 2.9	1.3%	0.02, 2.6
Peripheral vascular disease	1.0%	0.4, 1.5	0.6%	0.03, 1.2	1.0%	0, 2.2	0.4%	0, 1.1	0.0%	0, 0
Mental Health										
Depression	28.8%	24.7, 32.8	22.7%	17.0, 28.4	20.4%	15.0, 25.9	24.6%	17.1, 32.1	9.2%	4.6, 13.9
Anxiety	32.5%	30.7, 36.4	25.5%	19.7, 31.3	27.4%	21.7, 33.1	33.6%	26.0, 41.1	19.8%	14.6, 25.1
PTSD	8.9%	7.0, 10.9	9.3%	4.9, 13.8	6.7%	4.3, 9.1	11.0%	5.7, 16.4	2.0%	0.4, 3.7
Bipolar disorder	8.2%	6.2, 10.2	12.1%	9.8, 14.4	2.7%	1.0, 4.3	10.3%	4.9, 15.8	9.0%	9.0, 9.0
Psychotic disorders	6.5%	5.3, 7.8	12.7%	8.6, 16.9	5.2%	3.2, 7.2	7.0%	3.0, 11.1	7.0%	2.2, 11.7
Suicide attempt/ideation	6.9%	5.4, 8.4	6.8%	3.1, 10.5	11.3%	6.6, 16.0	12.3%	5.5, 19.1	11.6%	9.8, 13.4
Substance Use										
Opioid	13.3%	11.2, 15.4	17.1%	14.3, 19.9	16.1%	11.0, 21.2	10.8%	6.6, 14.9	7.2%	2.4, 11.9
Alcohol	12.0%	10.2, 13.9	22.6%	17.4, 27.8	18.5%	12.9, 24.1	11.2%	5.1, 17.3	11.2%	9.5, 12.9
Hallucinogens	0.1%	0.02, 0.3	0.3%	0.2, 0.5	0.0%	0, 0	0.2%	0, 0.5	0.0%	0, 0
Methamphetamine	20.4%	16.5, 24.3	8.8%	4.4, 13.2	21.5%	16.2, 26.7	17.6%	12.0, 23.3	13.2%	7.4, 18.9
Cocaine	2.6%	2.3, 3.9	15.6%	11.5, 19.8	1.1%	0, 2.3	3.5%	0, 7.1	0.3%	0, 0.7
Cannabis	6.2%	4.5, 7.9	9.9%	5.7, 14.1	6.6%	2.3, 10.9	9.1%	5.1, 13.1	4.5%	0.8, 8.2

Estimates adjusted for age and sex

Abbreviation(s): COPD, chronic obstructive pulmonary disease. CAD, coronary artery disease. AMI, acute myocardial infarction. PTSD, post-traumatic stress disorder

recent homelessness and incarceration have historically been challenging to generate, there is often minimal information to guide healthcare policy and delivery for these populations. As more states expand their Medicaid programs to cover housing resources and medical services during incarceration, data systems that can be used for program evaluation are critical. For example, dozens of states are in the process of expanding Medicaid to people in prisons and jails through Medicaid 1115 Reentry Waivers.^{24–26} The processes and estimates we outline here could be scaled to inform the

CMS-mandated evaluations of these programs. Importantly, the underlying data infrastructure for this work is built on an open-source, international common data model using linkage processes that can be used in a variety of contexts.²⁷

Our work has important limitations. This cohort only includes Minnesotans who had a medical encounter within one of the participating health systems during a 3-year period, although this represents over 95% of the Minnesota adult population based on 2022 US Census estimates. The specific health condition estimates may not generalize

to other states, though the general trends reported here are likely to be similar in different jurisdictions. Second, we capture health conditions that have been diagnosed in a medical record. This leads to conservative estimates of health conditions, particularly substance use disorders, given variable accuracy in ICD-10 coding.^{28–30} Future work among our group aims to deploy natural language processing to capture additional information from the free text of clinical documentation. Third, we use a privacy preserving record linkage to deduplicate and link external data sources to EHR data. Through this process we may slightly underreport homelessness and incarceration; our best estimate is around 5–10% underreported. This would make our prevalence estimates somewhat more conservative. Finally, our measures of incarceration and homelessness include any within the past year and do not differentiate between different levels of service use within HMIS.

CONCLUSION

In this report, we utilized linked EHR and administrative data to provide statewide estimates of physical health, mental health, and substance use conditions for Minnesotans with recent homelessness or criminal justice involvement. To our knowledge, these data represent the first EHR-based statewide health condition estimates among these populations. We highlight important disparities between people with recent homelessness and incarceration compared with the general population, particularly with regards to mental health and substance use conditions. The MNEHRC data model represents an efficient, collaborative approach to providing real-time public health data, including information on people with recent homelessness or incarceration.

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Data Availability All pertinent data and materials are available upon reasonable request.

Declarations

Ethical Approval This study was determined to be a public health surveillance project that did not require review from institutional review boards.

Consent to Participate Not applicable.

Consent to Publish Not applicable.

Competing Interests The authors declare that they have no competing or conflicts of interest.

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REFERENCES

1. Acosta JD, Chandra A, Yeung D, et al. What Data Should Be Included in a Modern Public Health Data System. *Big Data*. 2022;10(Suppl 1):S9–S14. <https://doi.org/10.1089/big.2022.0205>
2. Samet J, Brownson RC. Reimagining Public Health: Mapping A Path Forward. *Health Aff (Millwood)*. 2024;43(6):750–758. <https://doi.org/10.1377/hlthaff.2024.00007>
3. DeSalvo KB, Kadakia KT, Chokshi DA. Medicine's Role in Reimagining Public Health: Reuniting Panacea and Hygeia. *JAMA Health Forum*. 2021;2(12):e214051. <https://doi.org/10.1001/jamahealthforum.2021.4051>
4. DeSalvo KB. Public Health 3.0: A Call to Action for Public Health to Meet the Challenges of the 21st Century. *Prev Chronic Dis*. 2017;14. <https://doi.org/10.5888/pcd14.170017>
5. Health Resources & Services Administration. Reimagining Public Health Infrastructure and the Health Workforce for the 21st Century. Twentieth Annual Report to the Secretary of the U.S. Department of Health and Human Services and the U.S. Congress. Published online October 2021.
6. Bensken WP, Krieger NI, Berg KA, Einstadter D, Dalton JE, Perzynski AT. Health Status and Chronic Disease Burden of the Homeless Population: An Analysis of Two Decades of Multi-Institutional Electronic Medical Records. *J Health Care Poor Underserved*. 2021;32(3):1619–1634. <https://doi.org/10.1353/hpu.2021.0153>
7. Gutwinski S, Schreiter S, Deutscher K, Fazel S. The prevalence of mental disorders among homeless people in high-income countries: An updated systematic review and meta-regression analysis. *PLoS Med*. 2021;18(8). <https://doi.org/10.1371/journal.pmed.1003750>
8. Fazel S, Yoon IA, Hayes AJ. Substance use disorders in prisoners: an updated systematic review and meta-regression analysis in recently incarcerated men and women. *Addiction*. 2017;112(10):1725–1739. <https://doi.org/10.1111/add.13877>
9. Principles to Guide National Data Collection on the Health of Persons in the Criminal Justice System - PMC. Accessed June 4, 2024. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6505315/>
10. Ahalt C, Binswanger IA, Steinman M, Tulskey J, Williams BA. Confined to Ignorance: The Absence of Prisoner Information from Nationally

- Representative Health Data Sets. *J Gen Intern Med*. 2012;27(2):160-166. <https://doi.org/10.1007/s11606-011-1858-7>
11. **Shearer RD, Vickery KD, Bodurtha P, et al.** COVID-19 Vaccination Of People Experiencing Homelessness And Incarceration In Minnesota. *Health Aff (Millwood)*. 2022;41(6):846-852. <https://doi.org/10.1377/hlthaff.2021.02030>
 12. **Khatrri UG, Howell BA, Winkelman TNA.** Medicaid Expansion Increased Medications For Opioid Use Disorder Among Adults Referred By Criminal Justice Agencies. *Health Aff (Millwood)*. 2021;40(4):562-570. <https://doi.org/10.1377/hlthaff.2020.01251>
 13. **Winkelman TNA, Margolis KL, Waring S, et al.** Minnesota Electronic Health Record Consortium COVID-19 Project: Informing Pandemic Response Through Statewide Collaboration Using Observational Data. *Public Health Rep*. 2022;137(2):263-271. <https://doi.org/10.1177/00333549211061317>
 14. Home. Chronic Conditions Data Warehouse. Accessed June 23, 2024. <https://www2.ccwdata.org>
 15. What is HMIS? Minnesota's HMIS. Accessed September 19, 2024. <https://www.hmismn.org/what-is-hmis>
 16. USDA ERS - Rural-Urban Commuting Area Codes. Accessed June 6, 2024. <https://www.ers.usda.gov/data-products/rural-urban-commuting-area-codes.aspx>
 17. Bureau UC. American Community Survey 2015–2019 5-Year Data Release. Census.gov. Accessed June 6, 2024. <https://www.census.gov/newsroom/press-kits/2020/acs-5-year.html>
 18. **Olson M, Schlafer RJ, Bodurtha P, Watkins J, Hougham C, Winkelman TNA.** Health profiles and racial disparities among individuals on probation in Hennepin County, Minnesota, 2016: a cross-sectional study. *BMJ Open*. 2021;11(9):e047930. <https://doi.org/10.1136/bmjopen-2020-047930>
 19. **Duncan A, Sanders N, Schiff M, Winkelman TNA.** Adaptations to jail-based buprenorphine treatment during the COVID-19 pandemic. *J Subst Abuse Treat*. 2021;121:108161. <https://doi.org/10.1016/j.jsat.2020.108161>
 20. Minnesota Department of Health Center of Excellence on Public Health and Homelessness. Minnesota Homeless Mortality Report, 2017–2021. Published online January 26, 2023:1–14.
 21. **Hawks L, Wang EA, Howell B, et al.** Health Status and Health Care Utilization of US Adults Under Probation: 2015–2018. *Am J Public Health*. 2020;110(9):1411-1417. <https://doi.org/10.2105/AJPH.2020.305777>
 22. **Barry R, Anderson J, Tran L, et al.** Prevalence of Mental Health Disorders Among Individuals Experiencing Homelessness: A Systematic Review and Meta-Analysis. *JAMA Psychiatry*. 2024;81(7):691-699. <https://doi.org/10.1001/jamapsychiatry.2024.0426>
 23. **Favril L, Rich JD, Hard J, Fazel S.** Mental and physical health morbidity among people in prisons: an umbrella review. *Lancet Public Health*. 2024;9(4):e250-e260. [https://doi.org/10.1016/S2468-2667\(24\)00023-9](https://doi.org/10.1016/S2468-2667(24)00023-9)
 24. A Roadmap to Evaluating 1115 Reentry Waiver Outcomes – Transitions Clinic. Accessed August 6, 2024. <https://transitionsclinic.org/policy-brief-1115-waiver/>
 25. HHS Releases New Guidance to Encourage States to Apply for New Medicaid Reentry Section 1115 Demonstration Opportunity to Increase Health Care for People Leaving Carceral Facilities | CMS. Accessed August 6, 2024. <https://www.cms.gov/newsroom/press-releases/hhs-releases-new-guidance-encourage-states-apply-new-medicaid-reentry-section-1115-demonstration>
 26. **Hinton E, Pillai A, Published AD.** Section 1115 Waiver Watch: Medicaid Pre-Release Services for People Who Are Incarcerated. KFF. August 19, 2024. Accessed October 28, 2024. <https://www.kff.org/other/issue-brief/section-1115-waiver-watch-medicaid-pre-release-services-for-people-who-are-incarcerated/>
 27. **Stang PE, Ryan PB, Racoosin JA, et al.** Advancing the science for active surveillance: rationale and design for the Observational Medical Outcomes Partnership. *Ann Intern Med*. 2010;153(9):600-606. <https://doi.org/10.7326/0003-4819-153-9-201011020-00010>
 28. **Guan H, Sundararajan V, Halfon P, et al.** Coding algorithms for defining comorbidities in ICD-9-CM and ICD-10 administrative data. *Med Care*. 2005;43(11):1130-1139. <https://doi.org/10.1097/01.mlr.0000182534.19832.83>
 29. **Kuang A, Xu C, Southern DA, Sandhu N, Guan H.** Validated administrative data based ICD-10 algorithms for chronic conditions: A systematic review. *J Epidemiol Popul Health*. 2024;72(4):202744. <https://doi.org/10.1016/j.jep.2024.202744>
 30. **Fiest KM, Jette N, Guan H, et al.** Systematic review and assessment of validated case definitions for depression in administrative data. *BMC Psychiatry*. 2014;14:289. <https://doi.org/10.1186/s12888-014-0289-5>

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